ShiftGuard10

Robust Image Classification

EE708 · Term Project

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Problem Statement

The Challenge

- Models assume train = test distribution
- Real-world images vary due to noise, blur, lighting, distortions
- Leads to distribution shift
- Results in poor performance on unseen data
- Constraint: No pretrained models

Why It Matters

- Real-world systems face unseen & unpredictable data
- Pretrained models may not generalize well
- Models often overfit training data
- Need to learn robust representations from scratch

Our Goal

- Compare ViT, Deep CNN, Hybrid CNN-MLP, Shallow CNN
- Train all models from scratch
- Use a consistent pipeline for fairness
- Evaluate generalization under distribution shift
- Identify the most robust architecture

Dataset & Data Understanding

37,000
Total Images

10
Classes

32×32
Image Size

29.4K / 7.6K
Train / Test Split

Preprocessing & Augmentation

- **Normalization**
  Mean/variance standardization per channel
- **Random Cropping**
  Introduces spatial variation during training
- **Horizontal Flipping**
  Improves translational invariance
- **Mixup Augmentation**
  Convex combo of samples; stronger regularization

Model Architectures

ViT

Vision Transformer

Global attention

- Patch embedding
- Multi-head attention
- Global receptive field
- No locality bias
- Needs warmup + reg.

ResNet-style

Deep CNN

Local + hierarchical

- Residual connections
- BatchNorm + ReLU
- Hierarchical features
- Strong spatial bias
- Good for local patterns

CNN + FC

Hybrid CNN-MLP

Spatial + classifier

- Conv feature extractor
- Flatten → MLP head
- Simple pipeline
- Loses spatial structure
- Limited expressivity

Lightweight

Shallow CNN

Fast, low capacity

- Fewer layers & params
- Fast training
- Low memory cost
- Struggles on complex patterns
- Weak under shift

Vision Transformer : Architecture Detail

Core Attention Mechanism

$$Attention(Q,K,V) = softmax(QK^T / \sqrt{d_k}) \cdot V$$

Key Advantages

- Global receptive field from layer 1
- Multi-head attention → diverse features
- Better long-range dependency modeling
- Superior robustness under shift

Training Challenges & Solutions

⚠ No Inductive Bias

✓ Apply strong data augmentation + Mixup

⚠ Unstable Attention (early)

✓ LR warmup gradually raises learning rate

⚠ Overfitting Risk

✓ Dropout regularization throughout layers

⚠ Slow Convergence

✓ Cosine annealing for smooth LR decay

Training Strategy

Optimizer
AdamW

Decouples weight decay from gradient updates — better generalization than vanilla Adam. Critical without pretraining.

LR Schedule
Cosine Annealing

Smooth transition from exploration to convergence. Avoids abrupt changes; helps find better loss minima.

Batching
Batch Size 128

Balances compute efficiency and gradient stability across all architectures for fair comparison.

Regularization
Dropout + Mixup

Dropout reduces co-adaptation. Mixup generates convex combinations of samples, boosting shift robustness.

ViT Stability
LR Warmup

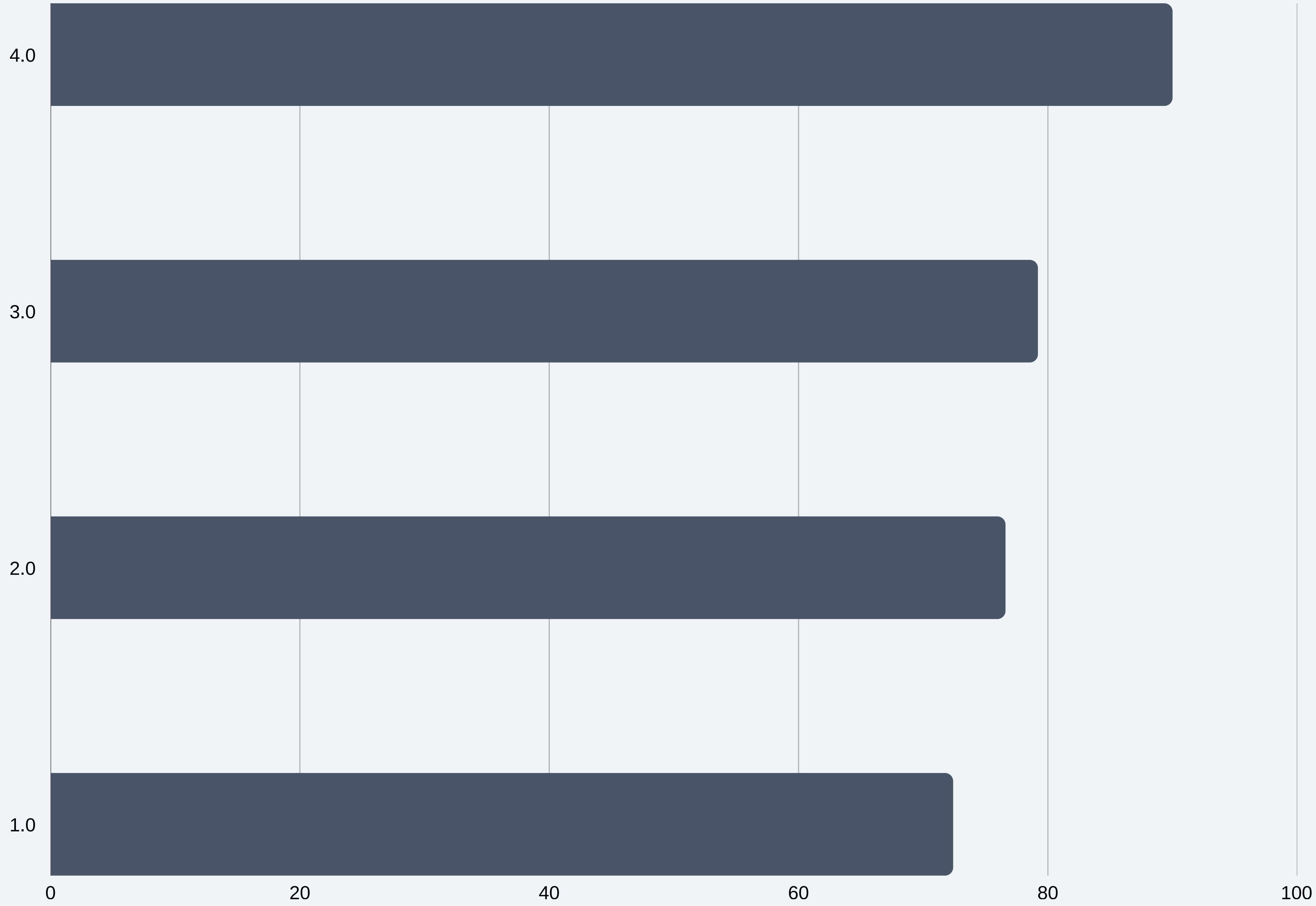
Gradually ramps LR from near-zero. Prevents unstable gradient updates in attention layers during early training.

Experimental Setup

All models trained on identical pipeline: isolating architecture as the only variable

Model	Architecture Settings
Vision Transformer (ViT)	Patch embedding, multi-head attn, LR warmup, dropout
Deep CNN (ResNet-style)	Stacked conv layers, BatchNorm, ReLU, residual links
Hybrid CNN-MLP	Conv feature extractor + Flatten + Fully-connected head
Shallow CNN	Fewer layers, reduced params, fast training

Results



	Acc.	F1	Pub.
Vision Transformer	90.0%	0.88	0.90
Deep CNN	79.2%	0.71	—
Hybrid CNN-MLP	76.6%	0.63	—
Shallow CNN	72.4%	0.58	—

Public Leaderboard F1: 0.90

Discussion

ViT Wins on Global Context

Transformers capture long-range dependencies from layer 1. Under distribution shift, global context matters more than local patterns — explaining the ~11% accuracy gap over Deep CNN.

CNN Strength: Locality, Weakness: Shift

Deep CNNs excel at local feature hierarchies but are limited to gradually expanding receptive fields. Without global attention, they struggle when distribution shifts affect non-local patterns.

Model Capacity Matters

Shallow CNN's 72.4% vs Deep CNN's 79.2% shows depth drives representational richness. However, deeper models require stronger regularization (dropout, weight decay) to avoid overfitting.

Training from Scratch Amplifies Differences

Without pretrained weights, inductive biases become decisive. ViT's lack of spatial priors, once a weakness, becomes a strength when compensated by scale and augmentation strategies.

Key Takeaways

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01 ViT Beats CNNs from Scratch

Vision Transformers achieve 90.0% accuracy (F1: 0.90 on public leaderboard), outperforming all CNN baselines by capturing global image context.

02 Global Attention = Shift Robustness

Multi-head self-attention provides holistic representations that are more stable under distribution shift than local convolutional filters.

03 Smart Training is Non-Negotiable

LR warmup, cosine annealing, AdamW, dropout, and Mixup are essential when training without pretrained weights as they close the gap vs. fine-tuning.

04 Depth > Breadth for CNNs

Deep CNN (79.2%) substantially beats Shallow CNN (72.4%), confirming that representational depth remains crucial even for convolutional networks.

Thank You